

SSH Brute Force Incident Report

SOC Investigation Summary

Executive Summary

A controlled SSH password guessing attack was run against an Ubuntu server. Repeated failed authentication events were collected from the Linux authentication log and analyzed in Splunk. The investigation confirmed password guessing activity from one source. No successful authentication from that source was observed during the investigated window.

Incident Details

Activity	SSH password guessing
Log Source	/var/log/auth.log
Analysis Platform	Splunk Enterprise
Target	Ubuntu Server with SSH enabled
Attack Source	Kali Linux running Hydra
MITRE ATT&CK	T1110.001 (Password Guessing)

Investigation

Hydra generated repeated SSH authentication attempts against the Ubuntu server. The resulting authentication events were available in Splunk. The source IP was extracted from the raw events and the failed attempts were grouped by source.

The search identified one source responsible for repeated failed authentication attempts. The same authentication data was then checked for successful logins from that source after the failures.

Finding

SSH password guessing was observed. No successful authentication from the attacking source was observed during the investigated window.

Analyst Conclusion

The evidence supports an attempted credential attack. The available authentication data does not show a successful SSH login from the attacking source during the investigated period.

Recommended Response

- Validate whether the source is expected.
- Review successful authentication around the same period.
- Check the targeted accounts for suspicious activity.
- Block the source when the activity is confirmed as unauthorized.
- Create and tune a Splunk alert for repeated SSH authentication failures.

What I Would Improve

In the next version, I would group failed logins into short time windows and test the threshold against normal SSH activity. I would also correlate repeated failures with any later successful authentication from the same source. This would make the detection more useful during real investigations.